

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EE)

Nov/Dec 2012

EE-215 Linear Electronics

Date: 29.11.2012, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) What is pn junction ? Discuss the behavior of pn junction under forward and reverse biasing? [05]
- (b) i) In a transistor, collector-base junction is always _____ biased. [02]
- ii) Zener diode operates in _____ region.
- Q - 2 (a) Explain working of full-wave bridge rectifier. Write their advantages and disadvantages. [06]
- (b) What is zener diode? Draw the equivalent circuit of an ideal zener. [06]
- (c) Why is zener diode used as a voltage regulator? [02]

OR

- Q - 2 (a) What is ripple factor? Derive its value for half-wave and full-wave rectifier. [06]
- (b) A half-wave rectifier is used to supply of 50V d.c. to a resistive load of 800Ω . The diode has a resistance of 25Ω . Calculate a.c. voltage required. [06]
- (c) Write the advantages of FETs [02]
- Q - 3 (a) Draw and explain the input and output characteristics of CB connection. [06]
- (b) Define β and show that $\beta = \alpha / 1 - \alpha$. [04]
- (c) The four diodes used in a bridge rectifier circuit have forward resistances which may be considered constant at 1Ω and infinite reverse resistance. The alternating supply voltage is 240 V r.m.s. and load resistance is 480Ω . Calculate (i) mean load current and (ii) power dissipated in each diode. [04]

OR

- Q - 3 (a) Establish the following relations: [06]
- (i) $I_E = (\beta + 1) I_B + (\beta + 1) I_{CBO}$ (ii) $I_C = \beta I_B + I_{CEO}$
- (b) Explain i) Choke input filter ii) π - filter. [04]
- (c) Calculate I_E in a transistor for which $\beta = 50$ and $I_B = 20\mu A$. [04]

SECTION - II

- Q - 4 (a) Explain voltage divider bias method for transistor. [05]
(b) Write the application of MOSFET. [02]

- Q - 5 (a) Explain h-parameters of a transistor and its performance in h-parameters. [05]
(b) A transistor used in CE arrangement has the following set of h parameters when the d.c. operating point is $V_{CE} = 10$ volts and $I_C = 1$ mA. [05]

$$h_{ie} = 2000\Omega, h_{oe} = 10^{-4} \text{ mho}, h_{re} = 10^{-3}, h_{fe} = 50$$

Determine (i) input impedance (ii) current gain and (iii) voltage gain. The a.c. load seed by the transistor is $r_L = 600\Omega$.

- (c) Write a short notes on the following (i) d.c. load line (ii) operating point. [04]

OR

- Q - 5 (a) Give the difference between JFET and BJT. [05]

- (b) Fill in the blanks by inserting appropriate words [05]

- (i) common collector arrangement is generally used for _____.
(ii) Operating point means _____.
(iii) The best method of bias is _____.
(iv) The noise level in JFET is _____ as compared to ordinary transistor.
(v) The h parameter h_{re} is called _____.

- (c) what are the drawbacks of h parameter approach in the design of a transistor ? [04]

- Q - 6 (a) Explain the construction and working of MOSFET [05]

- (b) Define the JFET parameter and establish relation between them. [05]

- (c) A transistor uses potential divider method of biasing. $R_1 = 10K\Omega$, $R_2 = 5K\Omega$, $R_C = 1K\Omega$, $R_E = 2K\Omega$, $V_{CC} = 15V$. Draw the d.c. load line and determine the operating point. Assume the transistor to be of silicon. [04]

OR

- Q - 6 (a) Draw the h-parameter equivalent circuit of a linear circuit. [05]

- (b) A transistor uses potential divider method of biasing The operating point is chosen such that $I_C = 2$ mA, $V_{CE} = 3V$. If $R_C = 2.2K\Omega$, $V_{CC} = 9V$ and $\beta = 50$. Determine the values of R_1 , R_2 and R_E . Take $V_{BE} = 0.3V$ and $I_1 = 10I_B$. [05]

- (c) Explain cut-off, saturation point and active region in a transistor with diagram. [04]
